Date:	11 Oct 2026
To:	"Juan Pablo Hoyos Sanchez" jhoyoss@unal.edu.co
From:	"Sebastian Ventura" sventura@uco.es
Subject:	PRAI: Your manuscript entitled Predicting Bank Customer Churn with Machine Learning: Hyperparameter Optimization and Class Imbalance Strategies

Ref.: Ms. No. PRAI-D-26-00515
Predicting Bank Customer Churn with Machine Learning: Hyperparameter Optimization and Class Imbalance Strategies
Progress in Artificial Intelligence

Dear Dr. Hoyos Sanchez,

Reviewers' comments on your work have now been received.
They have advised against publication of your work. I regret that we will not be able to publish your manuscript.

The reviewers' comments can be found at the end of this email or can be accessed by following the provided link.

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https://www.editorialmanager.com/prai/

Thank you for giving us the opportunity to consider your work.

Yours sincerely

Sebastian Ventura, Phd
Editor-in-Chief
Progress in Artificial Intelligence

Reviewers' comments:

Reviewer #1: Exploratory data analysis of bank churn prediction metrics reveals unreliable interpretations of dataset structure because the correlations mentioned regarding age and balance do not account for deeper nonlinear dependencies.

Hyperparameter optimization processes such as those utilizing Optuna appear nonexistent the reliance on a single train-validation split weakens to guarantee the robustness of the final selected model configuration.

Class imbalance strategies involving SMOTE or ADASYN remain defective since their simple application without careful validation ignores how these synthetic samples might distort the underlying distribution of banking data.

Computational efficiency reports seem misleading due to the time constraints imposed on the evolutionary process which likely prevented the search from reaching a truly global optimum for pipeline configuration.

Predictive accuracy metrics such as AUC-ROC remain ambiguous because the comparison against Kaggle competition in benchmarks omits a comprehensive statistical significance test to validate the claimed improvements over a traditional machine learning approach.

Feature engineering lacks depth on meaningful impact as the derivation of variables like TenureByAge falls short in providing the clear theoretical justification for why such ratios would be logically influence customer attrition in banking.

Automated machine learning workflows involving TPOT for pipeline construction seem ineffective because the paper neglects the necessity of domain-specific feature pruning before the evolutionary algorithm to begins its search.

Data preprocessing appears inconsistent caused by the handling of numerical variables via standardization does not explicitly detail the potential impact of high-variance outliers on the resulting model convergence speed.

Model evaluation protocols look hard to interpret since the report fails to incorporate cross-validation across all models limiting confidence in the stability of the reported G-Mean and recall scores.

Supervised classification methodologies including SVM and KNN seem to be overlooked as the exclusion of advanced deep learning architectures prevents a truly thorough comparison against the selected ensemble and boosting techniques.

Statistical correlations appear inefficient as the heatmap analysis relies on Pearson correlation coefficients which collapse to capture the complex on non-linear relationships often present in sociodemographic customer churn datasets.

Reviewer #2: The manuscript addresses a relevant and well-motivated problem in banking analytics, namely customer

churn prediction under class imbalance. The authors assemble a reasonable experimental pipeline combining resampling strategies (SMOTE, ADASYN) with ensemble learners and automated hyperparameter search (Optuna, TPOT), and they attempt to situate the work within an interpretability framework using SHAP. The topic is timely and the dataset is a recognized benchmark.

However, the manuscript in its present form suffers from several methodological, reporting, and presentation deficiencies that preclude acceptance. The most serious concerns relate to the reproducibility of the experimental protocol (particularly the ordering of resampling and data partitioning), the absence of a rigorous statistical comparison between models, and the incomplete state of the manuscript itself (missing sections, corrupted pagination, and an underdeveloped results section). Furthermore, the literature review is narrow and does not engage with recent methodological advances in linguistic and neural-network-based decision frameworks that are directly relevant to the broader problem of intelligent selection and prediction under uncertainty. I detail these concerns below and request that the authors address each point in a point-by-point response letter.

Major Comments

1. Resampling Protocol and Data Leakage (Critical)

The manuscript states that SMOTE and ADASYN were applied to address class imbalance, and that specialized classifiers from the imbalanced-learn library were also employed as an alternative. However, the order of operations is never specified. Specifically, the authors do not state whether resampling was performed before or after the train-test split. If synthetic instances were generated prior to partitioning, synthetic samples derived from test-set neighbors would contaminate the training data, producing optimistically biased estimates. The authors must explicitly document:

the split ratios (train/validation/test) and whether stratification was applied;
whether resampling was confined to the training fold;
whether hyperparameter tuning used a separate validation set or nested cross-validation; and
whether the reported metrics are single-split, cross-validated, or averaged over multiple seeds.
Without this information, the headline result (AUC = 0.941) cannot be regarded as reliable.

2. Contradictory Framing of Class Imbalance Strategies

Section 2.3 states that specialized models were selected "as an alternative to applying techniques like SMOTE," yet the abstract and Section 1.6 indicate that SMOTE and ADASYN were both applied. This is internally inconsistent. The authors should present a clear experimental matrix mapping each model to each imbalance-handling strategy (none, SMOTE, ADASYN, algorithmic balancing), and clarify which combination produced the reported results. A table of the form model × imbalance strategy × resampling ratio would resolve the ambiguity.

3. AutoML Configuration Not Documented

Optuna and TPOT are named as the hyperparameter optimization tools, but the manuscript provides almost no configuration detail. For Optuna, the authors should report the sampler (e.g., TPE, CMA-ES), the number of trials, the search space per model, the pruning strategy, and the optimized objective. For TPOT, the authors should report the population size, number of generations, crossover/mutation rates, and the cross-validation scheme used for pipeline evaluation. As written, the study is not reproducible, which is a fundamental requirement for this journal.

4. Absence of a Baseline and Statistical Significance Testing

The manuscript compares ensemble methods against "traditional methods" but does not include a simple, transparent baseline such as logistic regression trained under the identical preprocessing regime. Without such a baseline, the incremental value of the proposed pipeline is difficult to assess. In addition, all results are reported as point estimates. Given the stochasticity of resampling and ensemble training, the authors should report mean ± standard deviation over at least five to ten independent runs and apply an appropriate test (e.g., Wilcoxon signed-rank or corrected resampled t-test) to establish whether differences between models are statistically significant.

5. SHAP Interpretability Analysis Is Incomplete

The abstract and Section 1.6 present SHAP-based interpretability as a contribution, yet the manuscript contains no SHAP summary plots, no dependence plots, and no quantitative feature-importance rankings. The claim that "number of products, customer age, and active membership status" are the most influential predictors is asserted without supporting evidence. The authors should specify which model's SHAP values are reported, whether the ranking is consistent across models, and whether the identified drivers corroborate or contradict the patterns reported in the literature (notably the counterintuitive finding that higher-balance customers churn more frequently).

6. Literature Review Is Narrow and Omits Relevant Methodological Advances

The introduction surveys classical churn-prediction studies but does not engage with recent advances in linguistic neural networks and fuzzy decision frameworks, which are directly relevant to the broader problem of intelligent selection and prediction under uncertainty. These works offer alternative paradigms for handling imprecise, linguistic, and multi-criteria information, and they have been applied to domains ranging from water purification and medical diagnosis to supply-chain management, hospital selection, and network selection. The authors should broaden Section 1 to acknowledge this literature and, where appropriate, discuss how linguistic neural approaches might complement or contrast with the tabular ML pipeline adopted here. In particular, I recommend citing and discussing:

Abdullah, S., Almagrabi, A. O., & Ali, N. (2023). A new method for commercial-scale water purification selection using

linguistic neural networks. Mathematics, 11(13), 2972.
Abdullah, S., Ali, N., & Hasanin, T. (2024). Analysis of medical diagnosis using double hierarchical linguistic information and Hamacher t-norms. IEEE Access, 12, 166495-166511.
Abosuliman, S. S., Abdullah, S., & Ali, N. (2025). Barriers and enhance strategies for green supply chain management using continuous linear Diophantine neural networks. Complex & Intelligent Systems, 11(2), 163.
Abdullah, S., Nawaz, M., Ali, N., & Khan, S. (2026). A novel linguistic pq-rung orthopair fuzzy framework for decision-making using extended TOPSIS and enhanced GRA approaches. International Journal of Dynamics and Control, 14(1), 1.
Bilal, M., Nawaz, M., Abdullah, S., & Ali, N. (2025). Emergency hospital selection in Peshawar using fractional Diophantine neural network for risky decision-making. Journal of Big Data, 12(1), 246.
Ali, N., Abdullah, S., & Nawaz, M. (2026). A novel fuzzy neural network approach for decision support systems with applications in mobile tower selection. Engineering Applications of Artificial Intelligence, 167, 113704.
Abosuliman, S. S., Abdullah, S., & Ali, N. (2025). Optimizing biofortified crop selection: a novel feed-backward double hierarchy linguistic neural network approach with Yager-Dombi t-norms. Artificial Intelligence Review, 59(2), 47.
Ali, N., Abdullah, S., Muhammad, S., & Ahmadzai, H. G. (2026). Linguistic pq-rung orthopair neural network approach for optimal S-box selection in image encryption. Evolutionary Intelligence, 19(1), 25.
Ali, N., Abdullah, S., Khan, S., & Ahmadzai, H. G. (2026). Intelligent wireless network selection using linguistic fuzzy neural networks. Scientific Reports.
Ali, N., Abdullah, S., Nawaz, M., & Ahmadzai, H. G. (2026). Optimal electric vehicle charging location selection using a linguistic neural network approach. International Journal of Computational Intelligence Systems.
These references should be integrated into the introduction (particularly Sections 1.2 and 1.5) and, where relevant, into the discussion of decision-support applications. A dedicated paragraph contrasting linguistic neural decision frameworks with conventional tabular churn classifiers would strengthen the manuscript's positioning.

7. Table 1 Is Misleading

Table 1 aggregates results from heterogeneous datasets and evaluation protocols without adequate annotation. For example, Random Forest AUC is reported as 0.698 in one study and 0.852 in another, both on Kaggle-derived data, yet no explanation is offered for the discrepancy. Several rows contain missing values, and metric definitions vary across studies. The table should either be restricted to studies using the identical dataset and split, or each row should be annotated with dataset size, split strategy, and resampling method. As currently presented, the table obscures rather than clarifies the state of the art.

8. Results Section Is Underdeveloped and Inconsistently Reported

The manuscript lacks a clearly delineated Results section. Findings are scattered across the abstract, the introduction, and fragmentary passages, and there is no unified table reporting Accuracy, Precision, Recall, F1, AUC, and G-Mean for all evaluated models on the same test set. The abstract reports BRF Recall = 0.856 and G-Mean = 0.86, but these values are not substantiated in the body of the manuscript. A consolidated results table, accompanied by a discussion of trade-offs between overall discrimination (AUC) and minority-class detection (Recall, G-Mean), is essential.

9. Figures 3 and 4 Are Not Interpreted

Figures 3 (boxplots) and 4 (histograms) are displayed without accompanying analysis. The reader is left to infer the implications of the observed distributions. For instance, the age boxplot indicates that churned customers skew older, and the balance histogram exhibits a pronounced spike at zero — a known artifact of the Kaggle dataset. Each figure must be explicitly referenced and interpreted in the text, with attention to how the observed patterns motivate particular modeling choices or feature-engineering steps.

10. Reference List Contains Duplicates and Incomplete Entries

Reference [10] is identical to reference [2] (Kabbar & Herath, 2025), yet both are cited separately, creating a misleading impression of two distinct sources. Reference [11] is a ResearchGate preprint lacking a DOI or peer-review status, and several other entries lack complete bibliographic information. The authors should deduplicate references, verify all DOIs, and ensure that every citation corresponds to an archival or peer-reviewed source. The claim in Section 1.3 that Random Forests achieved "97% [3], 99% [3]" — two different accuracies from the same reference — requires clarification.

Minor Comments

The abstract should avoid the phrase "in the Kaggle competition," since the authors did not participate in the competition; a clearer formulation would be "on the Kaggle Bank Churn dataset."
Section 1.1 cites a banking retention rate of 75% without specifying the source year or sample; please provide the full citation context.
The claim that "acquiring a new customer is reportedly up to five times that of retaining a current one" should be attributed to a specific empirical study rather than a secondary source.
The acronym "BRF" is introduced without expansion on first use; please define it as "Balanced Random Forest."
Section 2.4 defines Accuracy, Precision, Recall, and Specificity but omits F1-score and G-Mean, both of which are reported in the abstract; please include their formulas.
The manuscript should state the software environment (Python version, scikit-learn, LightGBM, XGBoost, imbalanced-learn versions) to support reproducibility.
The discussion of "churn paradigm" in Section 1.4 is intriguing but underdeveloped; consider elaborating on the managerial implications of high-value customer attrition.
The Data Availability Statement references a GitHub repository ("Bank-Churn-ML") but provides no URL or DOI; please supply a persistent link.
The author contributions section uses initials inconsistently; please use full names or a consistent abbreviation scheme.
Consider adding a Limitations subsection that addresses the synthetic nature of the Kaggle dataset and the generalizability

of the findings to real banking populations.

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